EXP 1 STEP 1-6

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

import warnings
warnings.filterwarnings('ignore')
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files telco.csv

**telco.csv**(text/csv) - 1932822 bytes, last modified: 8/7/2026 - 100% done

```
Saving telco.csv to telco.csv
```

```
import pandas as pd

df = pd.read_csv("telco.csv")
```

```
print(df.shape)
```

```
(7043, 50)
```

```
df.head()
```

| | Customer ID | Gender | Age | Under 30 | Senior Citizen | Married | Dependents | Number of Dependents | Country | State | ... |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 8779-QRDMV | Male | 78 | No | Yes | No | No | 0 | United States | California | ... |
| **1** | 7495-OOKFY | Female | 74 | No | Yes | Yes | Yes | 1 | United States | California | ... |
| **2** | 1658-BYGOY | Male | 71 | No | Yes | No | Yes | 3 | United States | California | ... |
| **3** | 4598-XLKNJ | Female | 78 | No | Yes | Yes | Yes | 1 | United States | California | ... |
| **4** | 4846-WHAFZ | Female | 80 | No | Yes | Yes | Yes | 1 | United States | California | ... |

5 rows × 50 columns

```
print(df.columns)
```

```
Index(['Customer ID', 'Gender', 'Age', 'Under 30', 'Senior Citizen', 'Married',
       'Dependents', 'Number of Dependents', 'Country', 'State', 'City',
       'Zip Code', 'Latitude', 'Longitude', 'Population', 'Quarter',
```

```
       'Referred a Friend', 'Number of Referrals', 'Tenure in Months', 'Offer',
       'Phone Service', 'Avg Monthly Long Distance Charges', 'Multiple Lines',
       'Internet Service', 'Internet Type', 'Avg Monthly GB Download',
       'Online Security', 'Online Backup', 'Device Protection Plan',
       'Premium Tech Support', 'Streaming TV', 'Streaming Movies',
       'Streaming Music', 'Unlimited Data', 'Contract', 'Paperless Billing',
       'Payment Method', 'Monthly Charge', 'Total Charges', 'Total Refunds',
       'Total Extra Data Charges', 'Total Long Distance Charges',
       'Total Revenue', 'Satisfaction Score', 'Customer Status', 'Churn Label',
       'Churn Score', 'CLTV', 'Churn Category', 'Churn Reason'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 50 columns):
 #   Column                             Non-Null Count  Dtype  
---  ------                             --------------  -----  
 0   Customer ID                        7043 non-null   object 
 1   Gender                             7043 non-null   object 
 2   Age                                7043 non-null   int64  
 3   Under 30                           7043 non-null   object 
 4   Senior Citizen                     7043 non-null   object 
 5   Married                            7043 non-null   object 
 6   Dependents                         7043 non-null   object 
 7   Number of Dependents               7043 non-null   int64  
 8   Country                            7043 non-null   object 
 9   State                              7043 non-null   object 
 10  City                               7043 non-null   object 
 11  Zip Code                           7043 non-null   int64  
 12  Latitude                           7043 non-null   float64
 13  Longitude                          7043 non-null   float64
 14  Population                         7043 non-null   int64  
 15  Quarter                            7043 non-null   object 
 16  Referred a Friend                  7043 non-null   object 
 17  Number of Referrals                7043 non-null   int64  
 18  Tenure in Months                   7043 non-null   int64  
 19  Offer                              3166 non-null   object 
 20  Phone Service                      7043 non-null   object 
 21  Avg Monthly Long Distance Charges  7043 non-null   float64
 22  Multiple Lines                     7043 non-null   object 
 23  Internet Service                   7043 non-null   object 
 24  Internet Type                      5517 non-null   object 
 25  Avg Monthly GB Download            7043 non-null   int64  
 26  Online Security                    7043 non-null   object 
 27  Online Backup                      7043 non-null   object 
 28  Device Protection Plan             7043 non-null   object 
 29  Premium Tech Support               7043 non-null   object 
 30  Streaming TV                       7043 non-null   object 
 31  Streaming Movies                   7043 non-null   object 
 32  Streaming Music                    7043 non-null   object 
 33  Unlimited Data                     7043 non-null   object 
 34  Contract                           7043 non-null   object 
 35  Paperless Billing                  7043 non-null   object 
 36  Payment Method                     7043 non-null   object 
 37  Monthly Charge                     7043 non-null   float64
 38  Total Charges                      7043 non-null   float64
 39  Total Refunds                      7043 non-null   float64
 40  Total Extra Data Charges           7043 non-null   int64  
 41  Total Long Distance Charges        7043 non-null   float64
 42  Total Revenue                      7043 non-null   float64
 43  Satisfaction Score                 7043 non-null   int64  
 44  Customer Status                    7043 non-null   object 
 45  Churn Label                        7043 non-null   object 
 46  Churn Score                        7043 non-null   int64  
 47  CLTV                               7043 non-null   int64  
 48  Churn Category                     1869 non-null   object 
 49  Churn Reason                       1869 non-null   object 
dtypes: float64(8), int64(11), object(31)
memory usage: 2.7+ MB
```

```
df.describe(include='all')
```

| | Customer ID | Gender | Age | Under 30 | Senior Citizen | Married | Dependents | Number of Dependents | Country | |
|---|---|---|---|---|---|---|---|---|---|---|
| **count** | 7043 | 7043 | 7043.000000 | 7043 | 7043 | 7043 | 7043 | 7043.000000 | 7043 | |
| **unique** | 7043 | 2 | NaN | 2 | 2 | 2 | 2 | NaN | 1 | |
| **top** | 3186-AJIEK | Male | NaN | No | No | No | No | NaN | United States | C |
| **freq** | 1 | 3555 | NaN | 5642 | 5901 | 3641 | 5416 | NaN | 7043 | |
| **mean** | NaN | NaN | 46.509726 | NaN | NaN | NaN | NaN | 0.468692 | NaN | |
| **std** | NaN | NaN | 16.750352 | NaN | NaN | NaN | NaN | 0.962802 | NaN | |
| **min** | NaN | NaN | 19.000000 | NaN | NaN | NaN | NaN | 0.000000 | NaN | |
| **25%** | NaN | NaN | 32.000000 | NaN | NaN | NaN | NaN | 0.000000 | NaN | |
| **50%** | NaN | NaN | 46.000000 | NaN | NaN | NaN | NaN | 0.000000 | NaN | |
| **75%** | NaN | NaN | 60.000000 | NaN | NaN | NaN | NaN | 0.000000 | NaN | |
| **max** | NaN | NaN | 80.000000 | NaN | NaN | NaN | NaN | 9.000000 | NaN | |

11 rows × 50 columns

```
df.isnull().sum()
```

| | 0 |
|---|---|
| **Customer ID** | 0 |
| **Gender** | 0 |
| **Age** | 0 |
| **Under 30** | 0 |
| **Senior Citizen** | 0 |
| **Married** | 0 |
| **Dependents** | 0 |
| **Number of Dependents** | 0 |
| **Country** | 0 |
| **State** | 0 |
| **City** | 0 |
| **Latitude** | 0 |
| **Population** | 0 |
| **Referred a Friend** | 0 |

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.drop_duplicates(inplace=True)
```

```
df['Total Charges'] = pd.to_numeric(df['Total Charges'], errors='coerce')
```

```
numerical_cols = ['Tenure in Months','Monthly Charge','Total Charges']

for col in numerical_cols:
    df[col].fillna(df[col].median(), inplace=True)
```

```
categorical_cols = df.select_dtypes(include='object').columns

for col in categorical_cols:
    df[col].fillna(df[col].mode()[0], inplace=True)
```

```
df.isnull().sum()
```

| | |
|---|---|
| **Avg Monthly GB Download** | 0 |
| **Online Security** | 0 |
| **Online Backup** | 0 |
| **Device Protection Plan** | 0 |
| **Premium Tech Support** | 0 |
| **Streaming TV** | 0 |
| **Streaming Movies** | 0 |
| **Streaming Music** | 0 |
| **Unlimited Data** | 0 |
| **Contract** | 0 |
| **Paperless Billing** | 0 |
| **Payment Method** | 0 |
| **Monthly Charge** | 0 |
| **Total Charges** | 0 |
| **Total Refunds** | 0 |

| | |
|---|---|
| **Total Extra Data Charges** | 0 |
| **Total Long Distance Charges** | 0 |
| **Total Revenue** | 0 |
| **Satisfaction Score** | 0 |
| **Customer Status** | 0 |
| **Churn Label** | 0 |
| **Churn Score** | 0 |
| **CLTV** | 0 |
| **Churn Category** | 5174 |
| **Churn Reason** | 5174 |

**dtype:** int64

| | 0 |
|---|---|
| **Customer ID** | 0 |
| **Gender** | 0 |
| **Age** | 0 |
| **Under 30** | 0 |
| **Senior Citizen** | 0 |
| **Married** | 0 |
| **Dependents** | 0 |
| **Number of Dependents** | 0 |
| **Country** | 0 |
| **State** | 0 |
| **City** | 0 |
| **Latitude** | 0 |

```
#encoding
df_original = df.copy()
```

```
le = LabelEncoder()

columns = [
    'Gender',
    'Married',
    'Dependents',
    'Contract',
    'Payment Method',
    'Internet Service',
    'Churn Label'
]

for col in columns:
    df[col] = le.fit_transform(df[col])
```

```
scaler = StandardScaler()

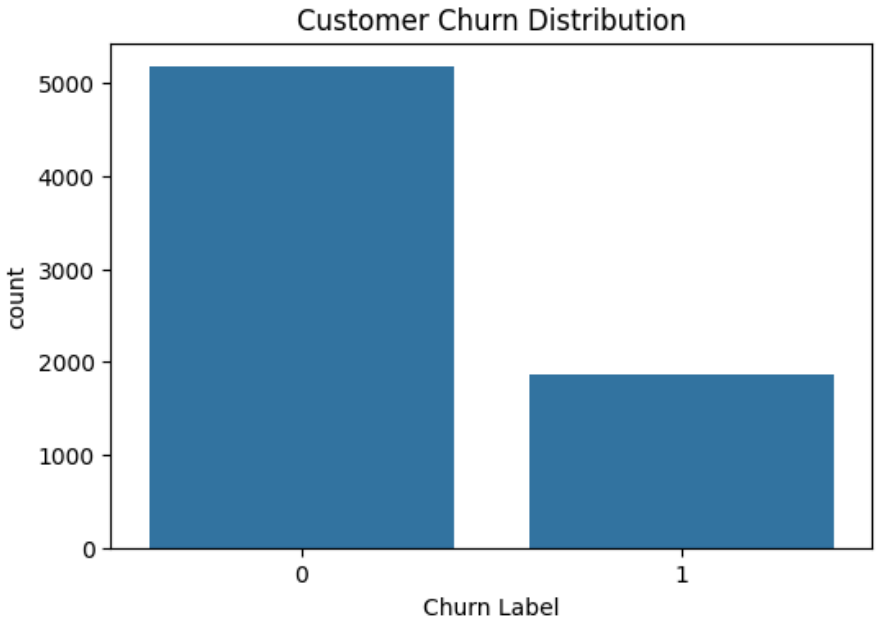
df[['Tenure in Months','Monthly Charge','Total Charges']] = scaler.fit_transform(
    df[['Tenure in Months','Monthly Charge','Total Charges']]
)
```

| | |
|---|---|
| **Online Security** | 0 |
| **Online Backup** | 0 |

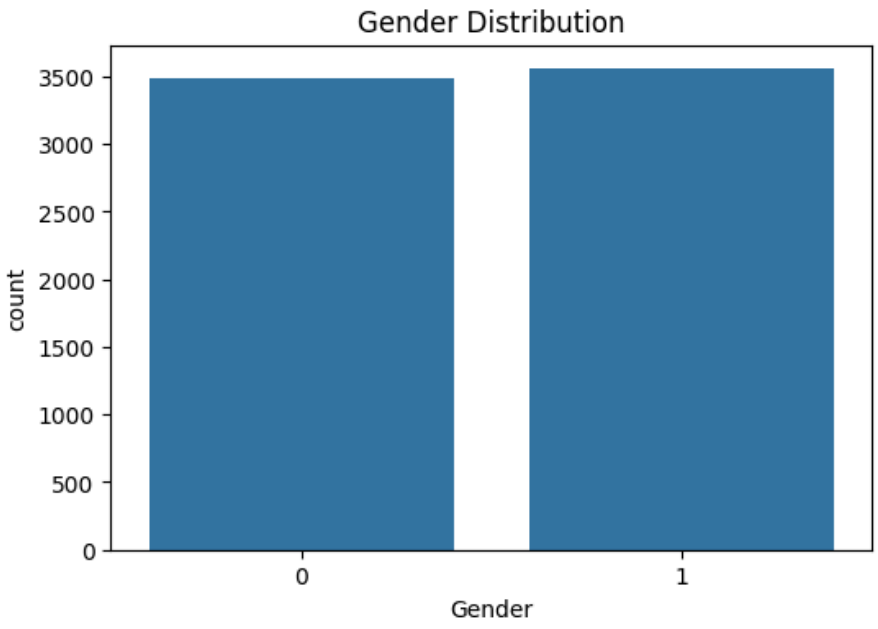
2. Analyze the distribution of: Customer Churn, Gender, Contract Type, Payment Method.

```
plt.figure(figsize=(6,4))
sns.countplot(x='Churn Label', data=df)
plt.title("Customer Churn Distribution")
plt.show()
```

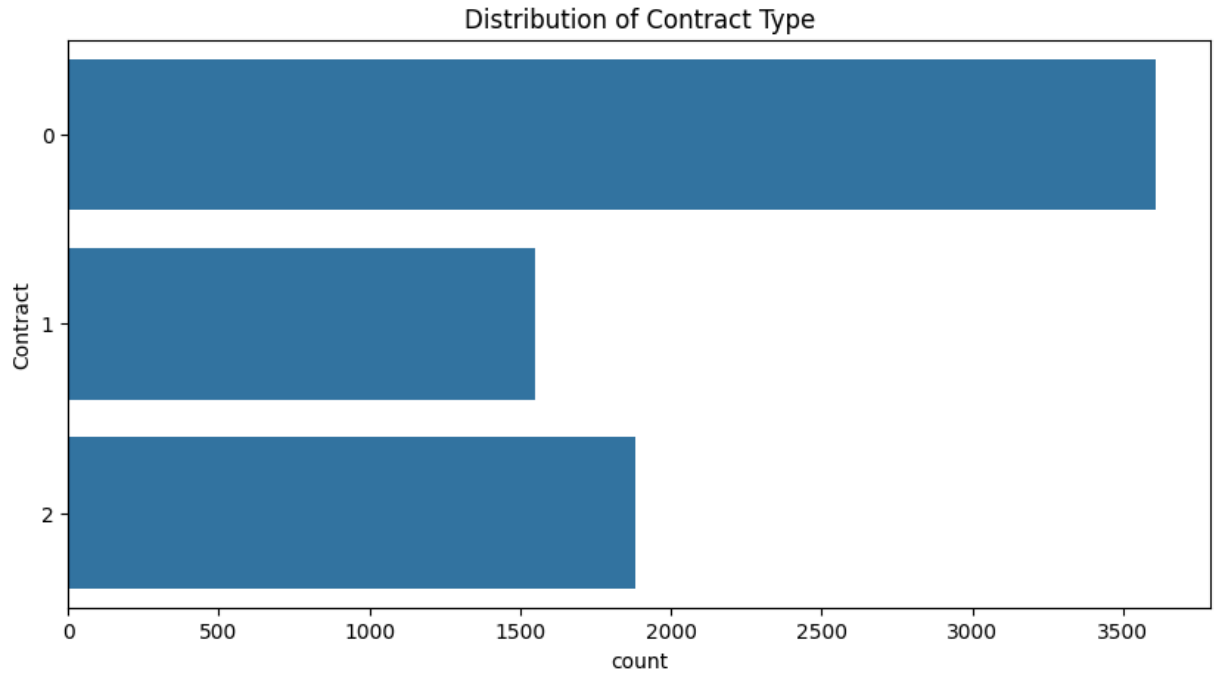
| | |
|---|---|
| **Streaming Movies** | 0 |
| **Streaming Music** | 0 |
| **Unlimited Data** | 0 |
| **Contract** | 0 |
| **Paperless Billing** | 0 |
| **Payment Method** | 0 |
| **Monthly Charge** | 0 |
| **Total Charges** | 0 |
| **Total Refunds** | 0 |



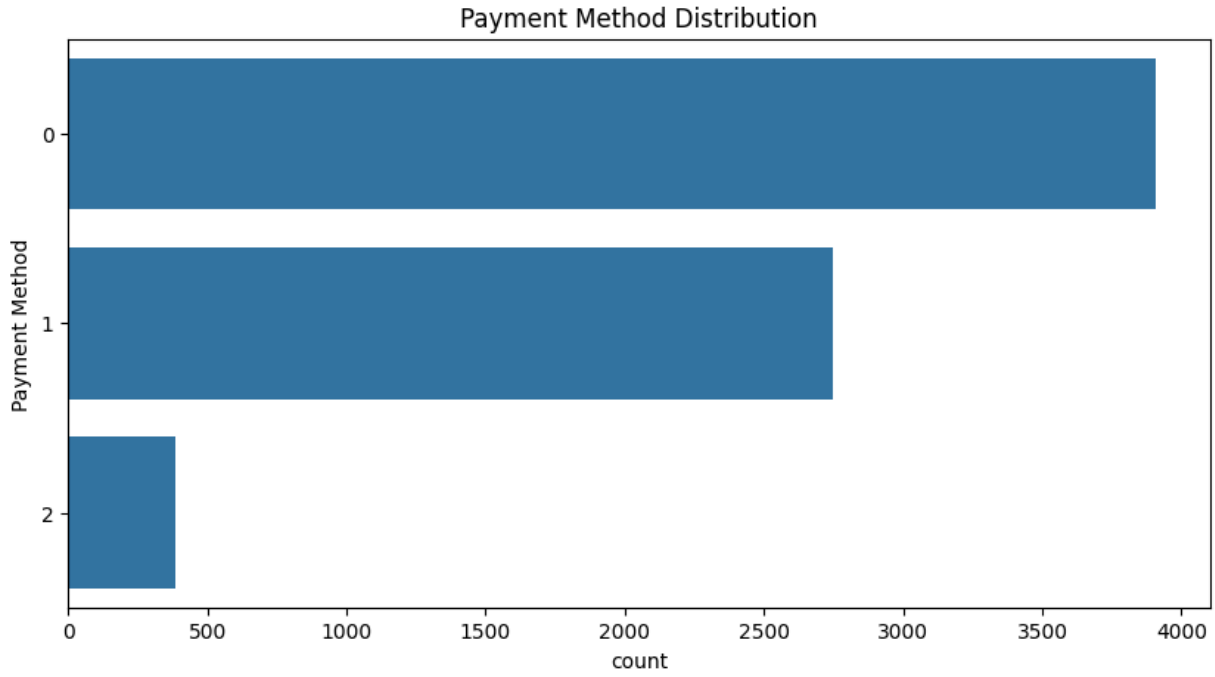

```
plt.figure(figsize=(6,4))
sns.countplot(x='Gender', data=df)
plt.title("Gender Distribution")
plt.show()
```




```
plt.figure(figsize=(10,5))
sns.countplot(y='Contract', data=df)
plt.title('Distribution of Contract Type')
plt.show()
```

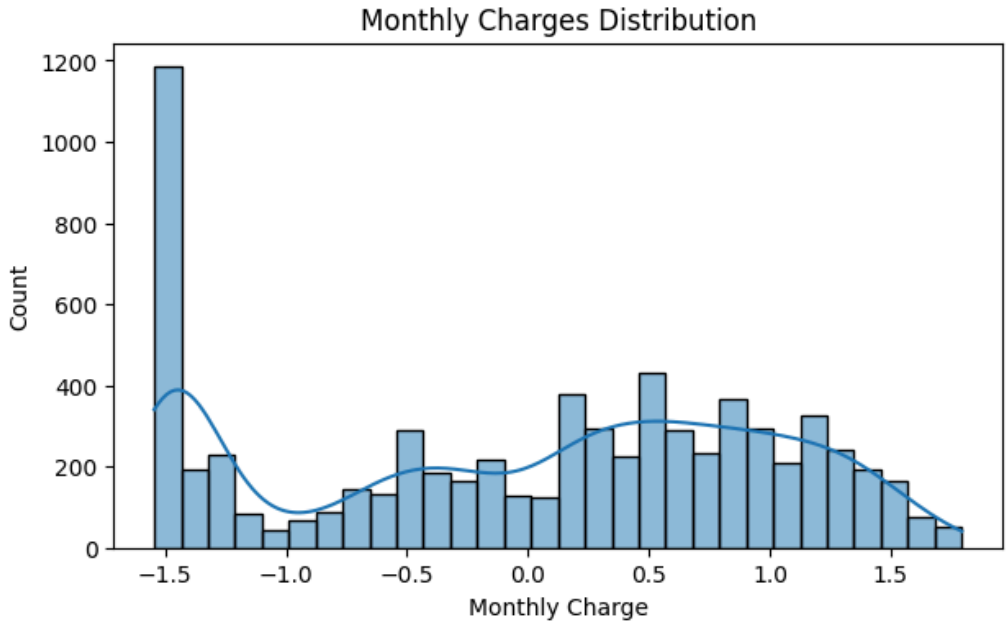



```
plt.figure(figsize=(10,5))
sns.countplot(y='Payment Method', data=df)
plt.title("Payment Method Distribution")
plt.show()
```

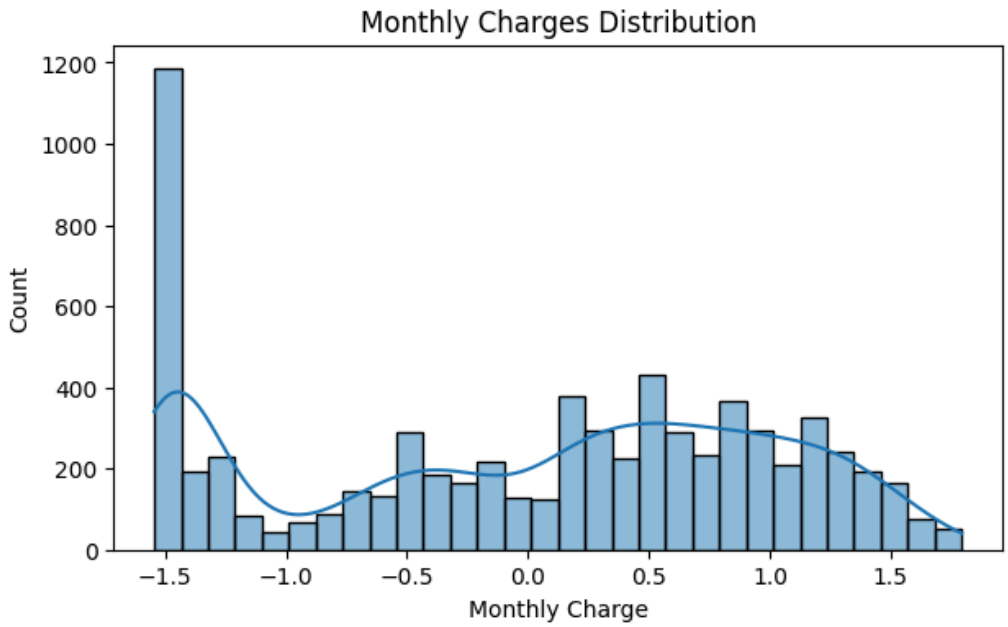



Task 3: Visualize MonthlyCharges, TotalCharges and Tenure

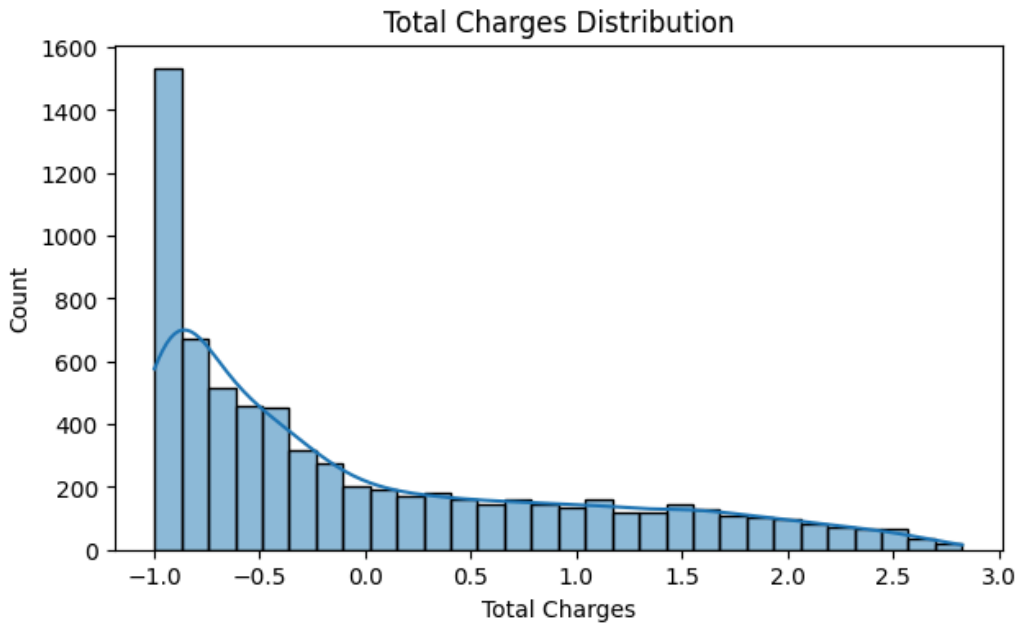
```
plt.figure(figsize=(7,4))
sns.histplot(df['Monthly Charge'], bins=30, kde=True)
plt.title("Monthly Charges Distribution")
plt.show()
```



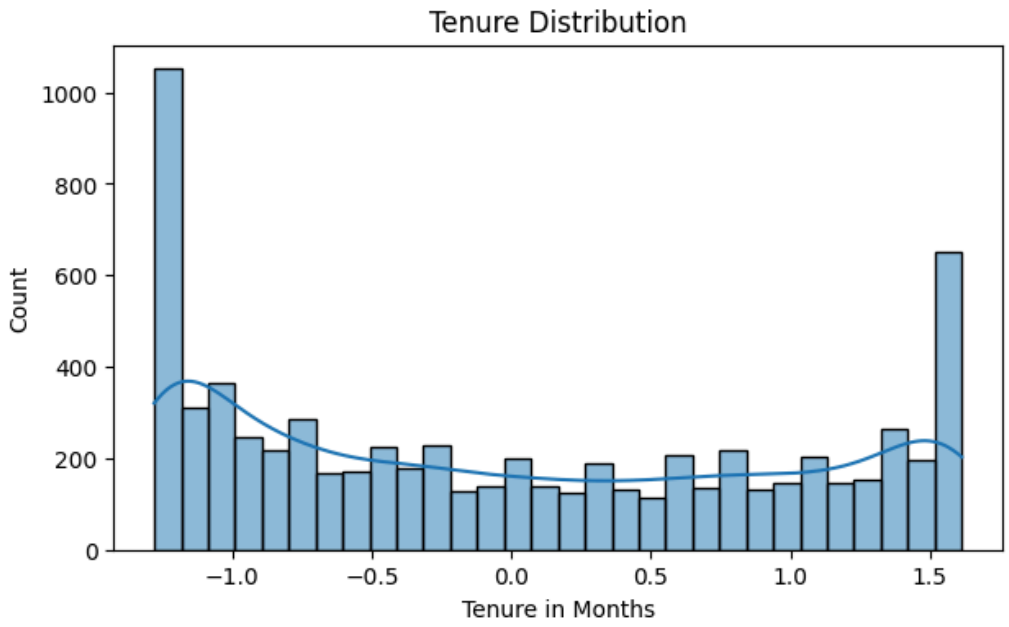

```
plt.figure(figsize=(7,4))
sns.histplot(df['Monthly Charge'], bins=30, kde=True)
plt.title("Monthly Charges Distribution")
plt.show()
```



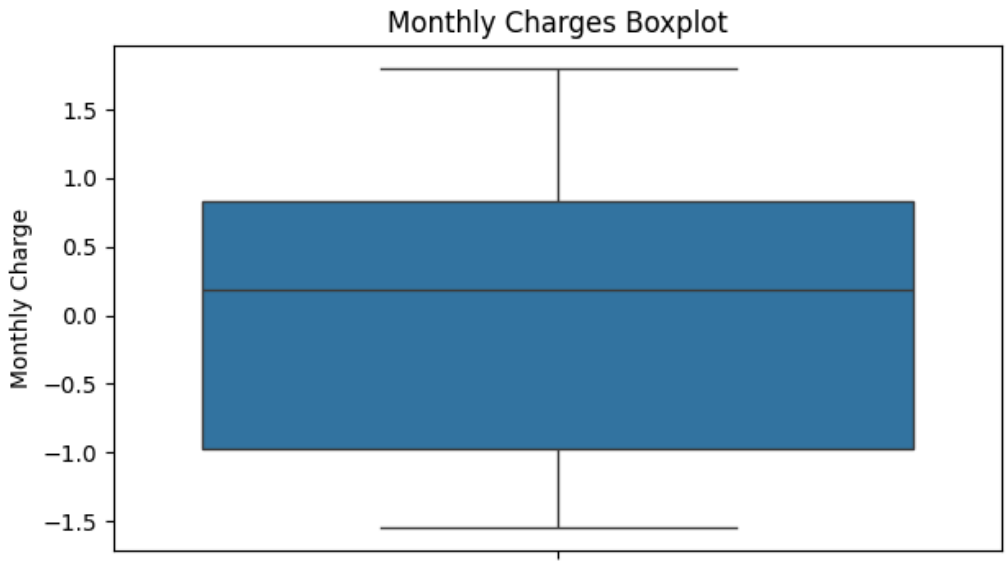

```
plt.figure(figsize=(7,4))
sns.histplot(df['Total Charges'], bins=30, kde=True)
plt.title("Total Charges Distribution")
plt.show()
```



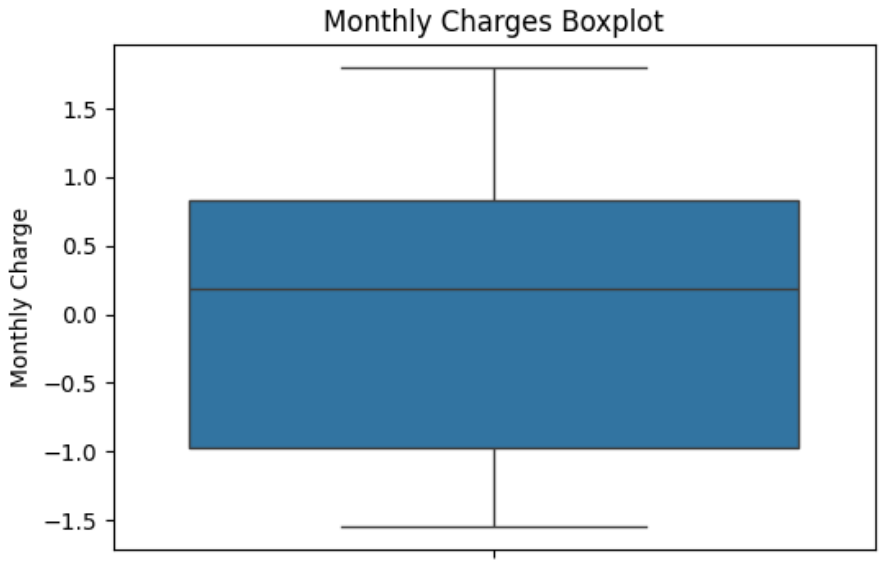

```
plt.figure(figsize=(7,4))
sns.histplot(df['Tenure in Months'], bins=30, kde=True)
plt.title("Tenure Distribution")
plt.show()
```



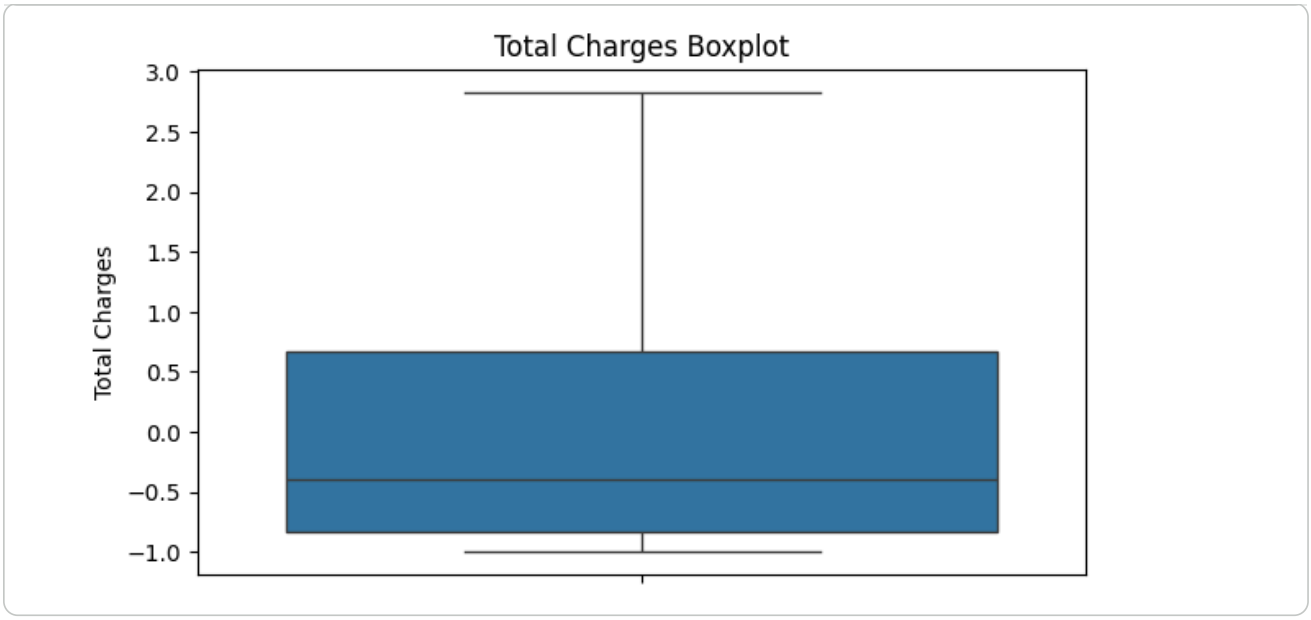

```
plt.figure(figsize=(7,4))
sns.boxplot(y='Monthly Charge', data=df)
plt.title("Monthly Charges Boxplot")
plt.show()
```



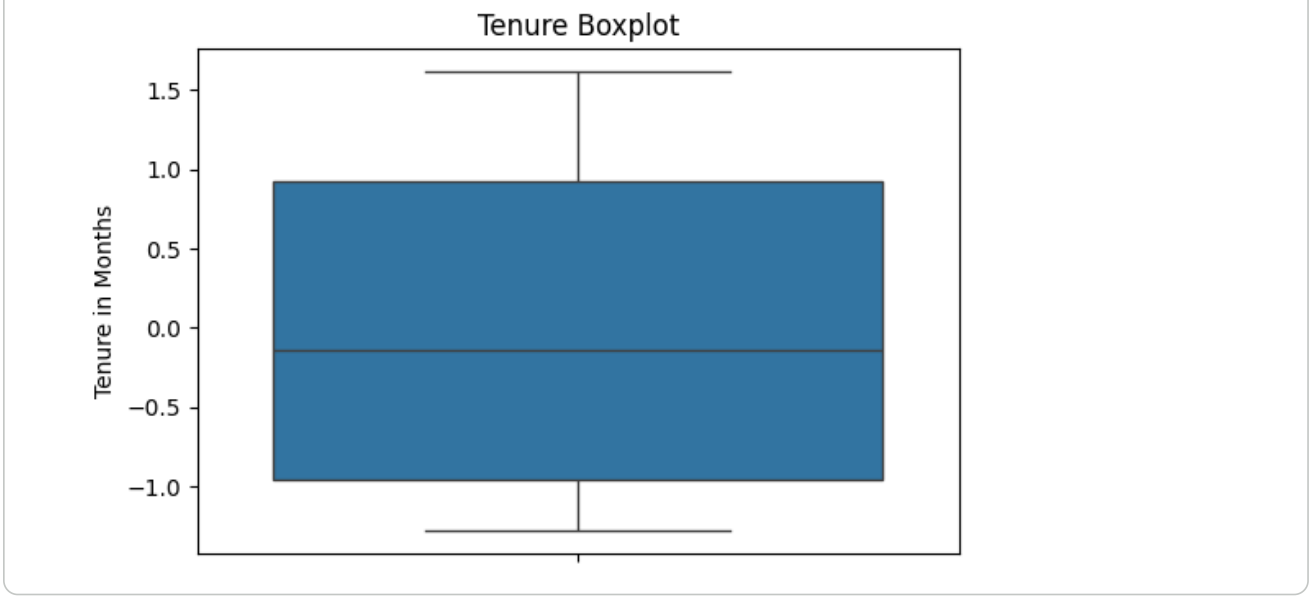

```
plt.figure(figsize=(6,4))
sns.boxplot(y=df['Monthly Charge'])
plt.title("Monthly Charges Boxplot")
plt.show()
```




```
plt.figure(figsize=(7,4))
sns.boxplot(y='Total Charges', data=df)
plt.title("Total Charges Boxplot")
plt.show()
```




```
plt.figure(figsize=(6,4))
sns.boxplot(y=df['Tenure in Months'])
plt.title("Tenure Boxplot")
plt.show()
```




Task 4: Detect Outliers using IQR

```
numerical_cols = ['Tenure in Months','Monthly Charge','Total Charges']

for col in numerical_cols:

    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower) | (df[col] > upper)]

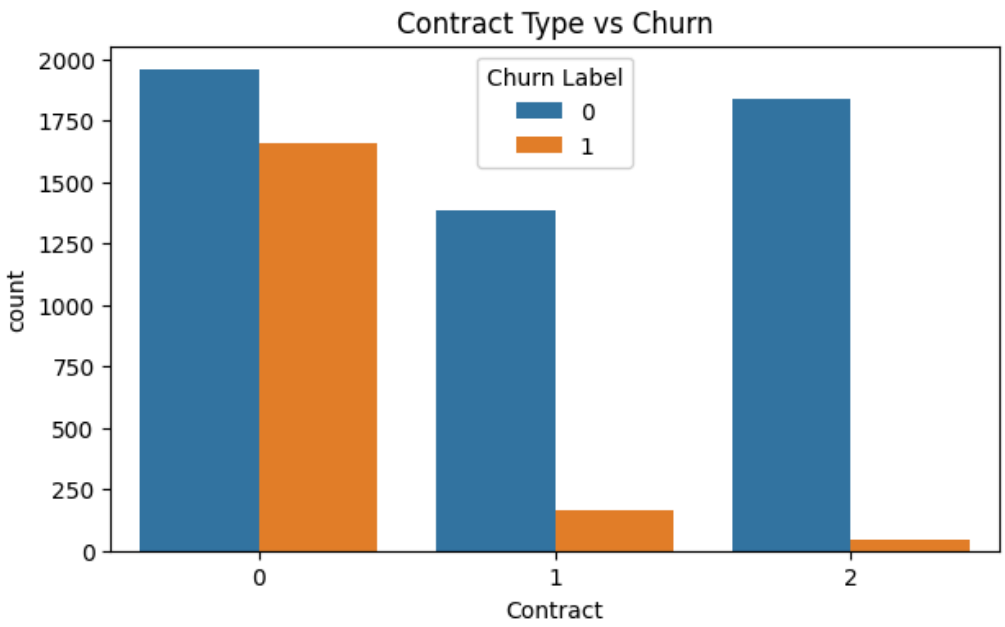
    print(f"{col} : {len(outliers)} outliers")
```

```
Tenure in Months : 0 outliers
Monthly Charge : 0 outliers
```

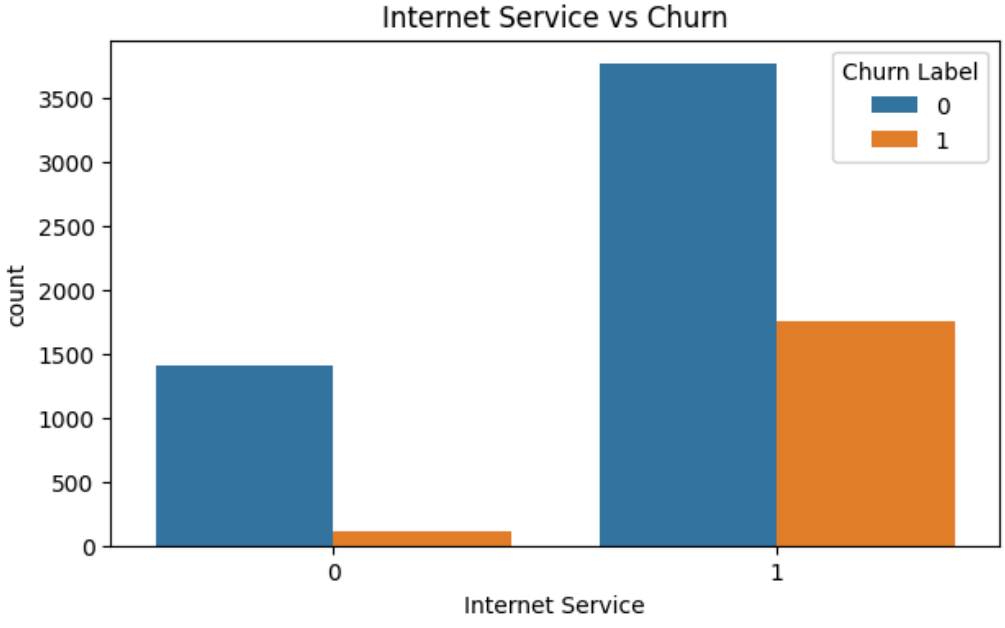
```
Total Charges : 0 outliers
```

Task 5: Relationship Analysis

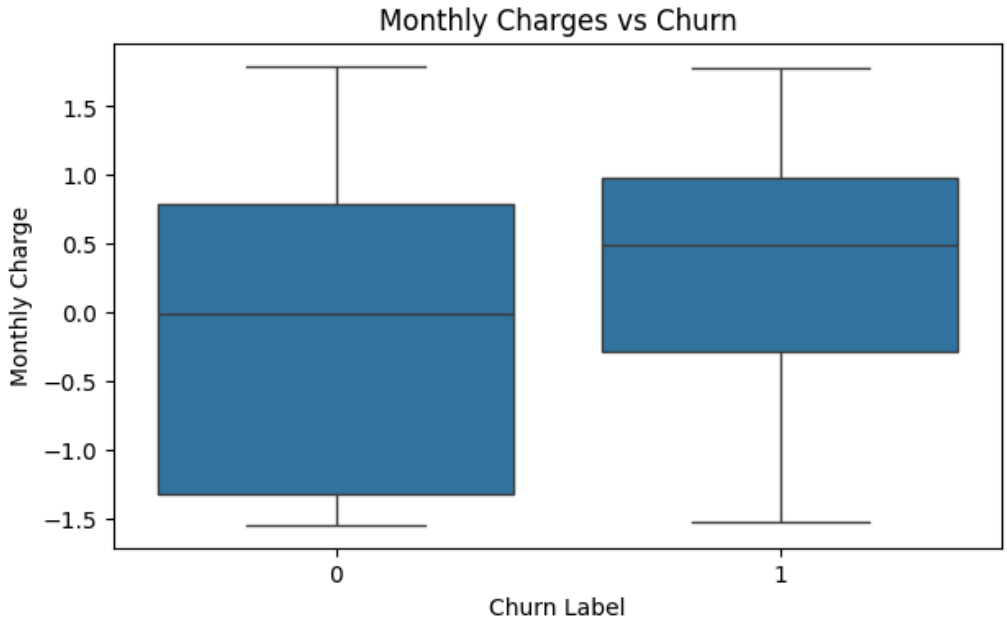
```
plt.figure(figsize=(7,4))
sns.countplot(x='Contract', hue='Churn Label', data=df)
plt.title("Contract Type vs Churn")
plt.show()
```



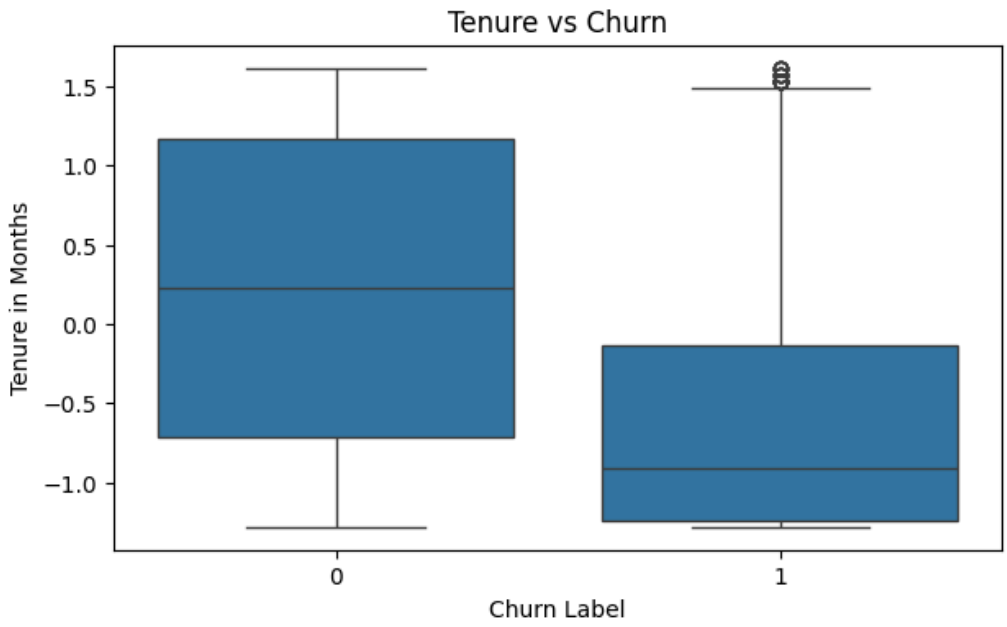

```
plt.figure(figsize=(7,4))
sns.countplot(x='Internet Service', hue='Churn Label', data=df)
plt.title("Internet Service vs Churn")
plt.show()
```




```
plt.figure(figsize=(7,4))
sns.boxplot(x='Churn Label', y='Monthly Charge', data=df)
plt.title("Monthly Charges vs Churn")
plt.show()
```




```
plt.figure(figsize=(7,4))
sns.boxplot(x='Churn Label', y='Tenure in Months', data=df)
plt.title("Tenure vs Churn")
plt.show()
```




Task 6: Analytical Questions

```
#Which contract type has the highest customer churn?
pd.crosstab(df['Contract'], df['Churn Label'])
```

| Churn Label | 0 | 1 |
|---|---|---|
| Contract | | |
| 0 | 1955 | 1655 |
| 1 | 1384 | 166 |
| 2 | 1835 | 48 |

```
#2. Which payment method is most frequently used by customers who churn?
df[df['Churn Label']==1]['Payment Method'].value_counts()
```